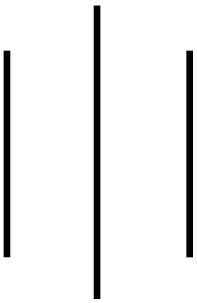
LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal

A Lab Report

Of

"Data Structure and Algorithm"



Submitted to:

Bachelor of Computer Application (BCA) Program

In partial fulfilment of the requirements for the degree of BCA under

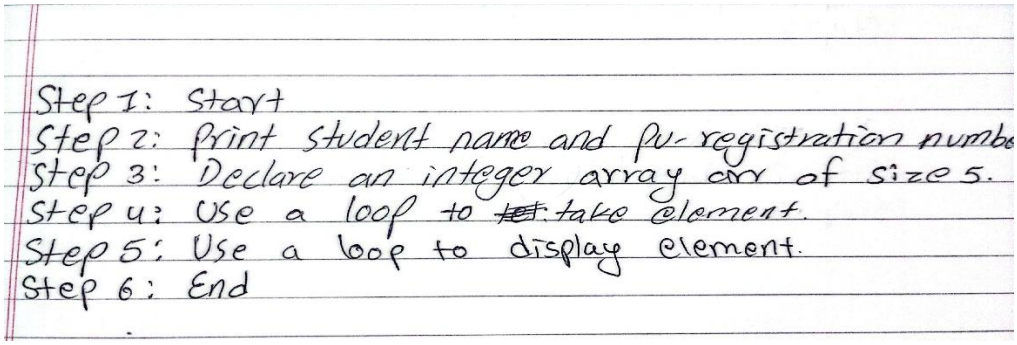
Pokhara University

Submitted by:

Name:	Course	Semester	P.U. Registration Number
Binay Lamichhane	BCA	3rd	2025-1-53-0319

Date: 12/10/2026

1. Write a C program to **input and display 5 elements** of an array.




```c
#include <stdio.h>
void main() {
    printf("Name:Binay Lamichhane \n");
    printf("Registration Number: 2025-1-53-0319 \n");
     int arr[5];
    int i;
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++) {
        scanf("%d", &arr[i]);
    }
    printf("The 5 elements are: ");
    for(i = 0; i < 5; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
```

Output

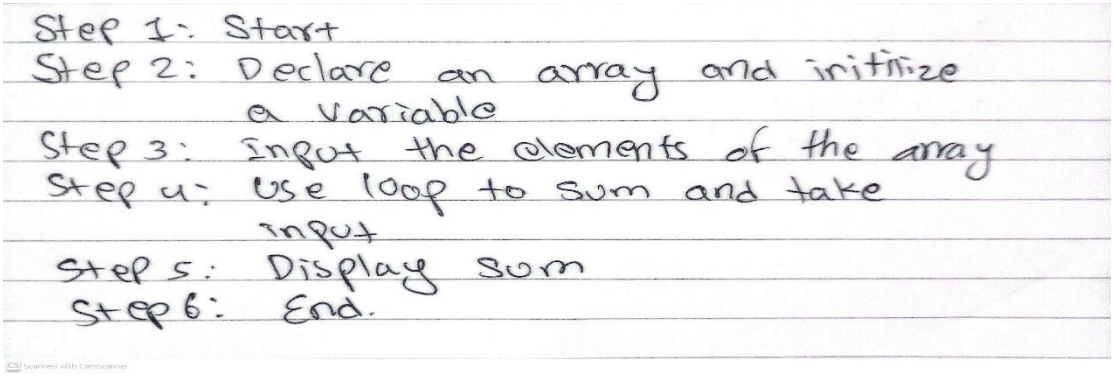
```
Name:Binay Lamichhane
Registration Number: 2025-1-53-0319
Enter 5 elements:
2
3
4
5
6
The 5 elements are: 2 3 4 5 6


=== Code Exited With Errors ===
```

Time Complexity: O(n)

Space Complexity: O(n)

2. Write a C program to **find the sum of all elements** in an array.



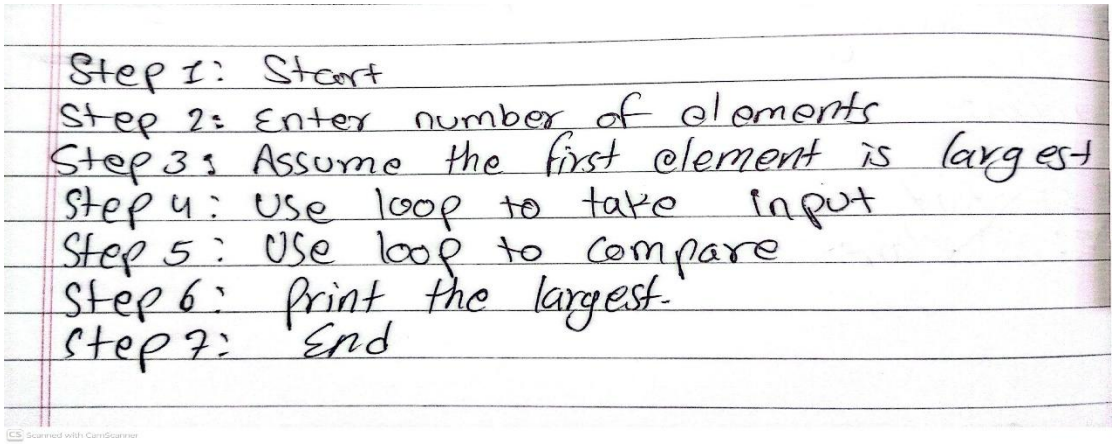

```c
#include <stdio.h>
void main() {
    printf("Name: Binay lamichhane \n");
    printf("Registration number: 2025-1-53-0319 \n ");
    int n, i, sum = 0;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }
    printf("Sum of all elements = %d\n", sum);
}
```

```
Name: Binay lamichhane
Registration number: 2025-1-53-0319
 Enter the number of elements: 3
Enter 3 elements:
4
5
7
Sum of all elements = 16
```

Time Complexity: O(n)

Space Complexity: O(n)

3. Write a C program to find the largest elements in the array.



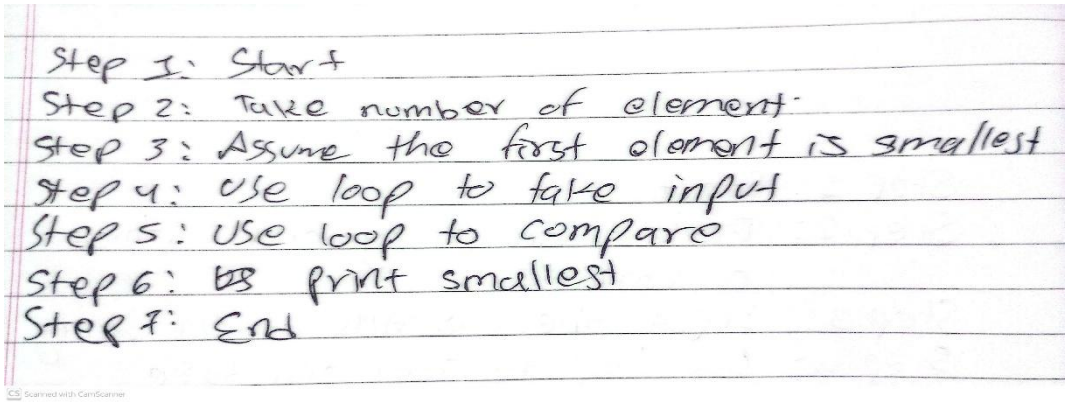

```c
#include <stdio.h>
void main() {
      printf("Name: Binay lamichhane \n");
     printf("Registration number: 2025-1-53-0319 \n ");
     int n, i;
     printf("Enter the number of elements: ");
     scanf("%d", &n);
     int arr[n];
     printf("Enter %d elements:\n", n);
     for(i = 0; i < n; i++) {
         scanf("%d", &arr[i]);
     }
     int max = arr[0];
     for(i = 1; i < n; i++) {
         if(arr[i] > max) {
             max = arr[i];
         }
     }
     printf("The largest element is: %d\n", max);
}
```

```
Name: Binay lamichhane
Registration number: 2025-1-53-0319
 Enter the number of elements: 3
Enter 3 elements:
55
77
44
The largest element is: 77
```

Time Complexity: O(n)

Space Complexity: O(n)

4. Write a C programming to find the smallest in array.



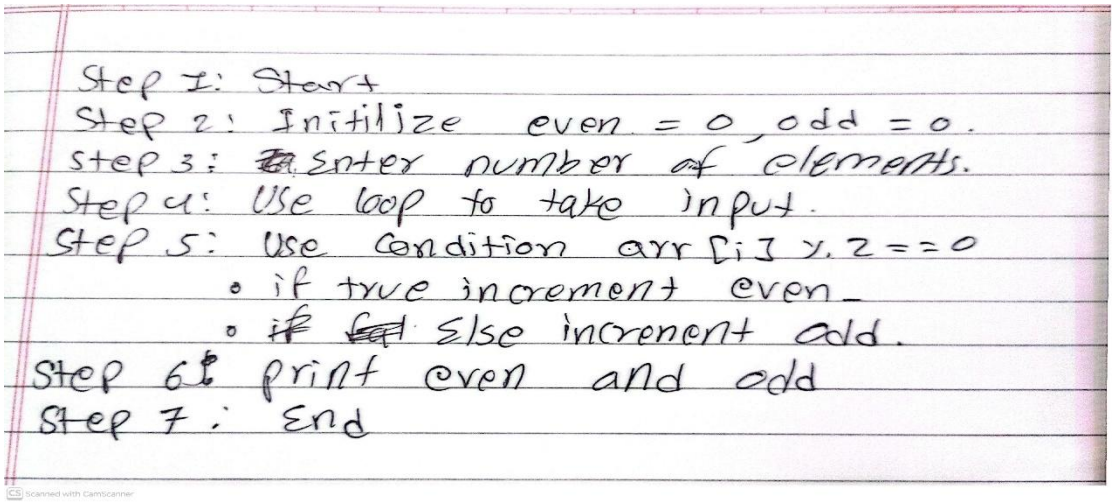

```c
#include <stdio.h>
void main() {
       printf("Name: Binay lamichhane \n");
     printf("Registration number: 2025-1-53-0319 \n ");
     int n, i;
     printf("Enter the number of elements: ");
     scanf("%d", &n);
     int arr[n];
     printf("Enter %d elements:\n", n);
     for(i = 0; i < n; i++) {
         scanf("%d", &arr[i]);
     }
     int min = arr[0];
     for(i = 1; i < n; i++) {
         if(arr[i] < min) {
             min = arr[i];
         }
     }
     printf("The smallest element is: %d\n", min);
}
```

```
Name: Binay lamichhane
Registration number: 2025-1-53-0319
 Enter the number of elements: 4
Enter 4 elements:
1
77
5
33
The smallest element is: 1
```

Time Complexity: O(n)

Space Complexity: O(n)

5. Write a C program to count the number of even and odd number in an array.




```c
#include <stdio.h>
void main() {
       printf("Name: Binay lamichhane \n");
      printf("Registration number: 2025-1-53-0319 \n ");
         int n, i;
      int even_count = 0, odd_count = 0;
      printf("Enter the number of elements: ");
      scanf("%d", &n);
      int arr[n];
      printf("Enter %d elements:\n", n);
      for(i = 0; i < n; i++) {
          scanf("%d", &arr[i]);
          if(arr[i] % 2 == 0) {
              even_count++;
          } else {
              odd_count++;
          }
      }
      printf("Number of even elements: %d\n", even_count);
      printf("Number of odd elements: %d\n", odd_count);
}
```

```
Name: Binay lamichhane
Registration number: 2025-1-53-0319
 Enter the number of elements: 5
Enter 5 elements:
3
4
5
1
7
Number of even elements: 1
Number of odd elements: 4
```

Time Complexity: O(n)

Space Complexity: O(n)